

Kavindu Ranathunga

Design Engineer, full-stack

I design and build web products end to end, from the interface people use to the API and AI behind it.

Experience

Sri Lanka Telecom Mobitel Full-Stack Developer Intern

Feb – Aug 2026

Top contributor to the TalentHub codebase. Internship extended by a month, then offered a freelance contract.

- **Owned TalentHub's UI/UX**, the platform that runs the company's trainee programme. Turned a non-responsive app into one mobile-ready design system for trainees and admins that new pages now follow.
- **Stopped fake daily logbook entries from 350+ interns**. Free local filters block gibberish, then Gemini catches off-topic entries. One batched AI call per entry instead of three kept it within API limits.
- **Cut admin report downloads from 2–3 minutes to under 30 seconds** on desktop and mobile.
- Rebuilt seat booking as a touch-friendly pan-and-zoom seat map that interns use every day.
- **Built OmniCatalog solo**, a tool for safely changing a product catalogue's structure, then recruited and mentored the intern team now carrying it forward.
- Built the EasyApply frontend solo, turning 9 paper forms into guided online forms in English, Sinhala and Tamil.

Projects

Zchema Next.js, TypeScript, PostgreSQL

github.com/kavindu-rakn/Zchema

Shows the blast radius of a database schema change before you run it, then applies, versions and rolls it back safely.

- Previews what a change would break (affected categories, record counts, values that can't be kept) instantly, using the same rules the database enforces.
- Previews never touch data and changes apply all-or-nothing, proven by tests that crash a migration halfway.

Luna React, Three.js

kavindu-rakn.github.io/Luna

A live moon-phase explorer with a 3D moon and astronomy-grade timings. Open source.

- Exact phase times and distances from Meeus' astronomical algorithms, with earthshine rendering, a time-travel calendar, timeline scrubbing and shareable links.
- Works for everyone: keyboard and screen-reader support, reduced motion, offline install, and a 3D engine that loads on demand.

Hotel Tamarind Tree Next.js, Prisma, PostgreSQL, Auth.js

github.com/kavindu-rakn/HotelTamarindTree

A direct-booking site for a working hotel, currently in review with its general manager.

- Three-step booking with live availability checks that prevent double-booking, an admin dashboard, and a nightly job that releases abandoned reservations.

Skills

Frontend	React, Next.js, TypeScript, Tailwind CSS, Framer Motion, GSAP, Three.js, accessibility, PWAs, i18n
Backend and data	Node.js, Express, REST APIs, PostgreSQL (PL/pgSQL, RLS), Supabase, Prisma, MongoDB
AI and workflow	Gemini API, Claude Code, Codex, Antigravity, Git, GitHub Actions, Vercel, automated testing

Education

Sri Lanka Institute of Information Technology (SLIIT)

Expected Sep 2027

BSc (Hons) Information Technology, specialising in Software Engineering

- **Final-year research:** Suvana, a two-way bridge between Deaf and hearing Sri Lankans. Building its gamified Sri Lankan Sign Language learning module for hearing users.